

Shiv Pratap Patel

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INTERNSHIP

CSIR CEERI, Pilani Rajasthan (Council of Scientific and Industrial Research)

July 2024 –
Dec 2024

Under Principal Scientist - Dr. Anil Kumar Saini (CSIR CEERI)

During my 6-months of IoT training at CSIR CEERI, I delved into a wide range of technologies and gained a proper understanding of IoT fundamentals. I explored various types of sensors and worked and learned their applications / limitations. I also became proficient in working with different technology stacks like Frontend , Backend , Database , Restful API, Fast API , Servers, Tunneling , Nodes and hardware components, such as microcontrollers, ESP32 , Arduino UNO , Raspberry Pi , and communication modules.

Rudra Robotics

July 2022 –
March 2023

After working for a year in a drone robotics company, I gained a comprehensive understanding of drone technology. I acquired hands-on experience in assembling and configuring drones, including the use of various components like motors, propellers, and flight controllers.

Also I have learned the drone simulation, utilizing software tools to test and optimize drone designs in virtual environments.

TECHNICAL SKILLS

Languages : JavaScript, Java basics, Python, PHP
Web Design : Tailwind, Bootstrap, HTML, Stylesheet (CSS)
Framework : React
APIs : FastAPIs, REST APIs
Database : NoSQL (MongoDB), SQL (MY SQL)
Development : VS Code , IntelliJ, Arduino IDE, VMware, Kali Linux(Basics) , Postman, MQTT Explorer
Tools : Firebase, JASON, Apache, XAMP, Mosquitto, HiveMQ, Node-RED basic
Miscellaneous: : Git, OpenAI, Data Structures and Algorithms, Problem Solving Skill
Robotics : Drone Simulation, Mechatronics
Blockchain : Bitcoin
AI & LLM Skills : Prompt Engineering for GenAI
Hardware Skills : ESP 32, Arduino UNO, HC-05, IOT, Circuit Designing

PROJECTS

• A Web Based Multi-Agent Thermal Comfort System

2024 | Frontend | Backend | APIs | | Iot
| Database | Hardwares | PHP |MQTT

Developed a web-based multi-agent thermal comfort system using Python (Flask) and JavaScript to optimize indoor climate conditions. The system leverages multi-agent algorithms for real-time data analysis and personalized comfort recommendations.

• Density Based Traffic Light System

Mini Project | 2022 | C Language

In this project I have developed a system through which the module will detect the density of traffic and accordingly it will manage the timer for red and green signal.

• Cafe's website

2023 | Python(Django)

Build a web application to manage a fictional cafe's online presence. Users can browse the menu, order items for delivery or pickup, and potentially make reservations. The cafe owner can manage orders, inventory, and potentially customer even information.

• House Price Predicting Application

2023 | Data Science | Python lib-Pandas, NumPy, scikit

This application uses data science to estimate the value of a property. You'll collect real estate data, clean and analyze it to identify key factors like square footage, bedrooms, and location that influence price. This can be a valuable both tool for buyers and sellers in the real estate market.

• Real Time Monitoring of govt buses using IOT

2024 | HTML 5| CSS | Python | C Language

Sensors like MQ-3 can monitor the vehicle's air for alcohol fumes exceeding a safe limit. Cameras (ESP 32) can track eye closure, blinking patterns, and head position to identify drowsiness. Alerts can include audible warnings, dashboard notifications, or vibration in the driver's seat.

• Robotic Car

2024 | C Language | HC-05

This project combines three functionalities: voice control for intuitive commands, object avoidance using UT sensors to navigate safely, and path following to autonomously traverse a predetermined course. Imagine a car responding to your voice instructions, stopping before bumping into objects, and all while following a specific path.

- **AMAZON & MYNTRA CLONE**

Developed a comprehensive e-commerce platform replicating the functionality of Amazon and Myntra. Leveraged HTML, CSS, and JavaScript to create a user-friendly interface with dynamic product listings

2024 | HTML | CSS | JS

COURSES AND CERTIFICATION

- ROBOTICS
- DATA SCIENCE
- AMD (FPGATHON)
- BLOCKCHAIN (NPTEL)
- DSA
- PYTHON
- FULL WEB STACK DEVOP.USING DJANGO
- COMMUNICATION
- DFIR (Digital Forensics & Incident Response)
- PROMPT ENGINEERING FOR GenAI

ACHIEVEMENTS (HACKATHON)

- 1st PRIZE - INTERNATIONAL HACKATHON CONDUCTED BY AMITY, Gr. NOIDA
- 1st PRIZE - NATIONAL HACKATHON CONDUCTED BY G.L BAJAJ, Gr.NOIDA
- 1st PRIZE – STATE LVL HACKATHON CONDUCTED BY LINGYAJ VIDYAPEETH, HARYANA
- 1st PRIZE – STATE LVL HACKATHON HACKATHON CONDUCTED BY:- NIET, Gr. NOIDA
- 2nd PRIZE – STATE LVL HACKATHON MANAV RACHNA FARIDABAD , HARYANA

PUBLICATION

- Density Based Traffic Control System Using IR Sensors and Microcontrollers:- Published in HBRP Publication Website :- www.hbrppublication.com 13th March 2023
- A Web Based Multi Agent-Thermal Comfort System:- In process for Amrita Vishwa Vidyapeetham IEEE Conference Jan 2025

PATENT APPLICATION, PUBLICATION

- Real Time Monitoring Of Govt. Buses Using IOT. Application No.202411076795 A 18 October 2024

EDUCATION

IIMT COLLEGE OF ENGINEERING, Greater Noida/(AKTU)
B. Tech. in Electronics and Communication Engineering

Greater Noida
2021 - 2025

B.B.S
12th CBSE – Science

Prayagraj
2019-2020

Naval Academy
10th CBSE

Gorakhpur
2017-2018

EXTRA-CURRICULAR ACTIVITIES

- President of Student's Council Cell, IIMT College Of Engineering Greater Noida
- Co-ordinator of Technical Club (INFERNO)

HOBBIES

- Playing Basketball and Chess